

CPU Performance & Frequency Test Report

System

- **Computer:** Apple iMac 11,3
- **Operating System:** Linux Mint Cinnamon
- **Kernel:** Linux 7.0.0-34-generic
- **CPU:** Intel Core i3-550
- **CPU architecture:** x86_64
- **Physical CPU cores:** 2
- **Threads:** 4
- **Base frequency:** 3.20 GHz
- **Detected maximum frequency:** 3.192 GHz
- **Detected minimum frequency:** 1.197 GHz
- **CPU frequency driver:** acpi-cpufreq

CPU Frequency Scaling

Linux initially reported the following CPU frequency governor:

```
scaling_governor = schedutil
```

The available governors were:

- conservative
- ondemand
- userspace
- powersave
- performance
- schedutil

For testing purposes, the governor was temporarily changed to:

```
performance
```

This does not overclock the processor. It simply instructs Linux to favor higher CPU performance states rather than aggressively reducing the frequency to save power.

Frequency Test

The following command was used to verify the maximum frequency:

```
cat /sys/devices/system/cpu/cpu0/cpufreq/scaling_cur_freq
```

During CPU load, the reported current frequency was:

3,192,000 kHz = 3.192 GHz

The configured maximum frequency was:

3,192,000 kHz = 3.192 GHz

Therefore, the processor successfully reached its configured maximum operating frequency.

Simple CPU Load Test

A simple Bash CPU workload was also performed:

```
time for i in {1..5000000}; do ;; done
```

The result was:

- **Real time:** 11.783 seconds
- **User CPU time:** 11.097 seconds
- **System CPU time:** 0.684 seconds

This test is not considered a proper CPU benchmark because Bash itself contributes significantly to the execution time. It was used only as a simple workload to observe CPU frequency behavior.

Conclusion

The Intel Core i3-550 is currently able to reach its configured maximum frequency of approximately **3.192 GHz** under load.

The temporary **performance** governor successfully allowed the CPU to operate at its maximum configured frequency. No actual overclocking was performed, and no CPU voltage or hardware-level frequency settings were modified.

The system therefore appears to be operating normally within the CPU frequency limits currently exposed by the Linux `acpi-cpufreq` driver.

Notes

The system is based on an older Apple iMac platform. Any attempt to increase the CPU frequency beyond the configured 3.192 GHz limit would require investigation of the specific Apple firmware, motherboard, CPU clock-generation hardware, thermal characteristics, and available low-level controls before making any changes.

The current test should therefore be regarded as a **CPU frequency and power-management experiment rather than an overclocking experiment.**